AI 101 for Mechanical Engineers · Module 13

What changes weights and what does not

Lesson 13.3

What you will learn

> Distinguish pretraining, fine-tuning, transfer learning, prompting and retrieval; identify which update weights; classify zero- and few-shot prompting; and separate weight updates from data-retention policy.

Lesson overview

Module 13 · Lesson 3

What changes weights and what does not

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1 Training: weights are updated

Pretraining
Learns broadly useful patterns from a large corpus.
updates weights

Fine-tuning
Further training on additional task examples.
updates weights

Transfer learning
Reuses earlier training for a new task: by fine-tuning, *or* with a frozen feature extractor whose weights stay fixed.
updates weights only if fine-tuned

2 Inference: context is supplied

Zero-shot prompt
Instructions and context, no demonstration examples.
no weight update

Few-shot prompt
Instructions plus one or more demonstration examples.
no weight update

Retrieval (RAG)
Adds a found passage, e.g. from a manual, to the context. A RAG system can still be fine-tuned as a separate step.
no weight update during the query

3 Classify the worked case

Two example lab summaries placed in the prompt	few-shot context
A retrieved pump-manual passage	retrieved context
Training on extra labelled examples	fine-tuning

Every route can still give a wrong engineering answer. Check the source, its revision and the result.

Not a privacy rule: "prompting does not update weights" describes the computation. It says nothing about what a provider stores or later trains on; check the provider's policy.

Key takeaway

Training changes weights; prompting and retrieval supply context.
Know which mechanism you used, and evaluate the output either way.

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Pretraining learns weights from broad data



- Weights: learned parameters used in the model’s computation
- Corpus: a collection of data used for training
- Learned patterns differ from instructions supplied later

Transfer learning may or may not change weights

Fine-tuning

- Additional examples update parameters [29]
- For a task, domain or behaviour

Frozen feature extractor

- Reuses the learned representation
- Extractor weights are not updated

> Both are transfer learning: reusing earlier training for another task [75].

Zero-shot and few-shot are both inference

Zero-shot

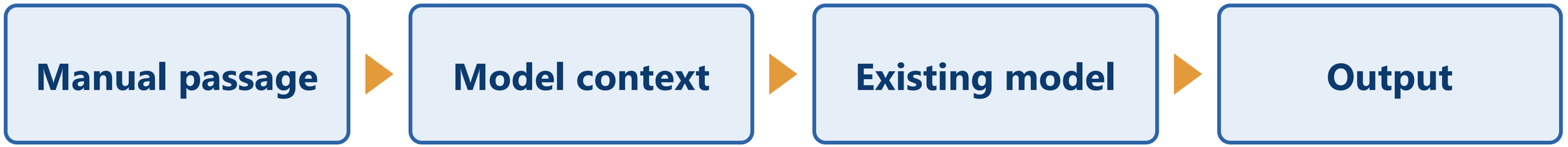
- No demonstration examples [67]

Few-shot

- Some examples in the prompt
- e.g. two example lab summaries

> Prompting supplies context during inference; it does not by itself update weights.

Retrieval supplies source passages as context



- RAG uses retrieved material to support generation [76]
- Retrieval during a query does not change weights
- A RAG system can be fine-tuned separately, as in [76]

Classify the three adaptation actions

Action	Mechanism	Updates weights?
Two example summaries in prompt	Few-shot inference context	Not by themselves
Retrieved manual passage	Source material in context	Not inherently
Training on labelled examples	Fine-tuning	Yes

> Each route still needs evaluation; first check the manual and its revision.

Weight updates and data policy are separate

Computation

- Prompting does not itself update weights

Provider policy

- Data retention is a separate matter
- Training policies are a separate question

> Do not infer privacy from the weight-update distinction.

CHECK YOUR UNDERSTANDING

Which changes weights: fine-tuning, adding examples to a prompt, or retrieving a manual?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Which changes weights: fine-tuning, adding examples to a prompt, or retrieving a manual?

> Fine-tuning updates weights through training. Prompt examples and a retrieved manual supply inference context without inherently updating the weights.

CHECK YOUR UNDERSTANDING

Can using a frozen feature extractor for another task count as transfer learning?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Can using a frozen feature extractor for another task count as transfer learning?

> Yes. Transfer learning reuses earlier training. It can use a frozen representation without weight updates, or include fine-tuning that changes parameters.

CHECK YOUR UNDERSTANDING

Does an absence of weight updates during prompting prove that submitted data are not retained?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does an absence of weight updates during prompting prove that submitted data are not retained?

> No. The computational operation and the provider's data-retention or separate training policy are different matters.

Key takeaways

- Pretraining and fine-tuning learn or update model parameters
- Transfer learning can include fine-tuning or a frozen feature extractor
- Zero-shot has no demonstrations; few-shot includes some
- Prompting and retrieval supply context without inherently retraining
- Evaluate every route; weight updates do not determine data retention

References

[29] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in Proc. 2019 Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol. (NAACL-HLT), 2019, pp. 4171–4186. doi: 10.18653/v1/N19-1423

[67] T. B. Brown et al., “Language models are few-shot learners,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.

[75] S. J. Pan and Q. Yang, “A survey on transfer learning,” IEEE Trans. Knowl. Data Eng., vol. 22, no. 10, pp. 1345–1359, 2010. doi: 10.1109/TKDE.2009.191

[76] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.

Numbers match the course reference list. The written lesson is the authoritative text.